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Geomorphometric attributes of the global system of rivers at 30-minute spatial resolution

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Abstract

In this paper we explore the geomorphometric characteristics and integrity of a 30′ (longitude × latitude) spatial resolution representation of the global system of potentially-flowing rivers. We quantify several geomorphometric attributes of digital, Simulated Topological Network (STN-30p) depicting potential flow pathways across the entire non-glacierized surface of the Earth. This data set was examined with respect to several metrics describing individual grid cells, river segments, and complete drainage systems. Nearly 60,000 grid cells constitute the global non-glacierized land mass. The cells are organized into more than 30,000 distinct river segments belonging to approximately 6200 drainage basins. STN-30p flow paths and drainage basins are classified as order one through six using the classification system of Strahler. STN-30p flow pathways depict rivers draining a global land area of 133.1×10^6 km^2. These pathways show a total length of 3.24×10^6 km at 30′ spatial resolution. The relationships between STN-30p order and interior river segment numbers, accumulated sub-basin areas, and accumulated length within individual basins yield high correlation coefficients (average $r^2 > 0.96$ for continents and globe). Mean values across individual continents and river orders for the bifurcation ratio (3.15 to 4.44), drainage area ratio (3.74 to 5.77), and basin length ratio (2.02 to 3.27) fall well within the ranges tabulated at finer spatial scales. A basin shape index, $S_b = L/A^{0.5}$, defined as a function of potential mainstem length and drainage area, varies between 1.0 and 5.0 for basins >25,000 km^2 and shows a global mean of 2.12. The structure of STN-30p potential river systems is consistent with those of rivers analyzed at finer spatial scales as demonstrated by the numerical similarity of the several geomorphometric indices analyzed. However, for a particular basin, indices from STN-30p will be based on a condensed set of river orders relative to those derived at finer scales. A first order STN-30p river is roughly equivalent to an order five-to-six river derived from 1:62,500 scale maps. While 30′ spatial resolution was found to represent well the 522 basins with areas >25,000 km^2 that drain 82% of the land mass, it cannot be used with high confidence in characterizing the geomorphometry of the remaining smaller basins. For global climate and biogeochemical studies, a composite of the 30′ resolution and finer spatial resolutions appears to be necessary. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Rivers; Drainage basins; Geomorphology; Hydrosphere

1. Introduction

The geomorphology of drainage basins and the organization of stream networks has been well-established for several decades (see Jarvis and

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Woldenburg, 1984). Quantitative tools emerged initially from field analysis of single, small catchments (e.g. Horton, 1945; Schumm, 1956) or of synthetic basins derived from statistical models (e.g. Shreve, 1966; Werner and Smart, 1973). Early regional-scale studies also exist, such as the summary of river network characteristics for the conterminous United States by Leopold et al. (1964). Recent research has focussed on scale-dependent extraction of drainage basin attributes (e.g. LaBarbera and Rosso, 1989; Lammers and Band, 1990; Helminger et al., 1993; Band and Moore, 1995) as well as assessments of the influence such attributes have on hydrological response (e.g. Beven et al., 1988; Band et al., 1991, 1995; Moore and Grayson, 1991; Famiglietti and Wood, 1994; Rodriguez-Iturbe, 1993; Sivapalan, 1993). However, these studies never progressed to the global scale and the generality of the statistics presented still requires testing as a precursor for use in global change studies.

We recently presented (Vörösmarty et al., 2000a) a gridded river networking scheme, global in domain and organized at 30′ spatial resolution and offered details on the construction and verification of this data base, its geographic co-registration to discharge and river chemistry monitoring stations, and an analysis of land-to-ocean linkages. We have applied versions of the STN-30p data set in water budget and river discharge studies at the regional (Vörösmarty et al., 1996a, 1991), continental (Lammers et al., 2000), and global scales (Fekete et al., 1999). It has also been used to study the impact of large reservoirs on continental runoff distortion and suspended sediment flux (Vörösmarty et al., 1997b,c). The 1° to 30′ scale is developing as the focal point for continental and global-scale water and constituent transport modeling (e.g. Seitzinger and Kroeze, 1998; Oki and Sud, 1998; Ludwig et al., 1996; Ludwig and Probst, 1998; Vörösmarty et al., 1997a–c; Oki et al. 1995; Esser and Kohlmaier, 1991), which will require simulated river networks like STN-30p. The 30′ spatial resolution appears to be a sensible compromise between the necessary level of topological detail and computational requirements of finer-scale global data sets (e.g. Graham et al., 1999; USGS-EDC, 1998). Ongoing work is aimed at developing tools to create and analyze the nature of aggregated river networks using finer-scale data sets.

2. Methods

The steps and algorithms used in constructing the digital river network data set are summarized in Fig. 1. We developed the Simulated Topological Network for potential flow pathways (STN-30p) by spatially aggregating to 30′ (longitude × latitude) the ETOPO5 five to ten-minute digital elevation model (DEM) (Edwards, 1989), which was the best global data set available to us at the time we initiated this study. Because the data set is in geographic coordinates, individual cell areas change with latitude (3091 km^2 at equator; 2176 km^2 at 45° latitude). The aggregated DEM was used to determine a maximum topographic gradient and a provisional direction of flow for each land-based grid cell (ARC/INFO; ESRI, Inc., Redlands CA). Inputs to each cell were assigned as single links from adjacent upstream pixels. Each directed link was given one of eight compass directions (N, NE, E, SE, S, SW, W, NW) (Burrough, 1986). A single direction of exit was assigned to each cell while inputs could be from any or all of the remaining directions.

A customized software product was employed (Global Hydrological Archive and Analysis System [GHAAS]; University of New Hampshire) when necessary to reconfigure the provisional network structure to align with digital overlays of rivers and independent map sources (see Vörösmarty et al. 2000a). We also made comparisons to available statistics associated with river monitoring stations, making corrections to the topology and/or checking the validity of the original source material to reconcile differences. We discovered several inconsistencies with respect to river length and upstream area in existing reports (Meybeck and Ragu 1995, 1997; Vörösmarty et al. 1996b; UNESCO 1965–84, 1995). These apparent errors and associated definitional problems are discussed at length in Vörösmarty et al. (2000a). Many such problems are associated with defining river courses and basin areas in presently arid regions where active runoff may occur in only a portion of the basin (e.g. Nile). At the global scale, median disparities between all STN-30p and previously reported basin areas in Vörösmarty et al. (1996b) and Meybeck and Ragu (Meybeck and Ragu, 1995, 1997) were 13 and 11%, respectively. The median disparity for river length compared to

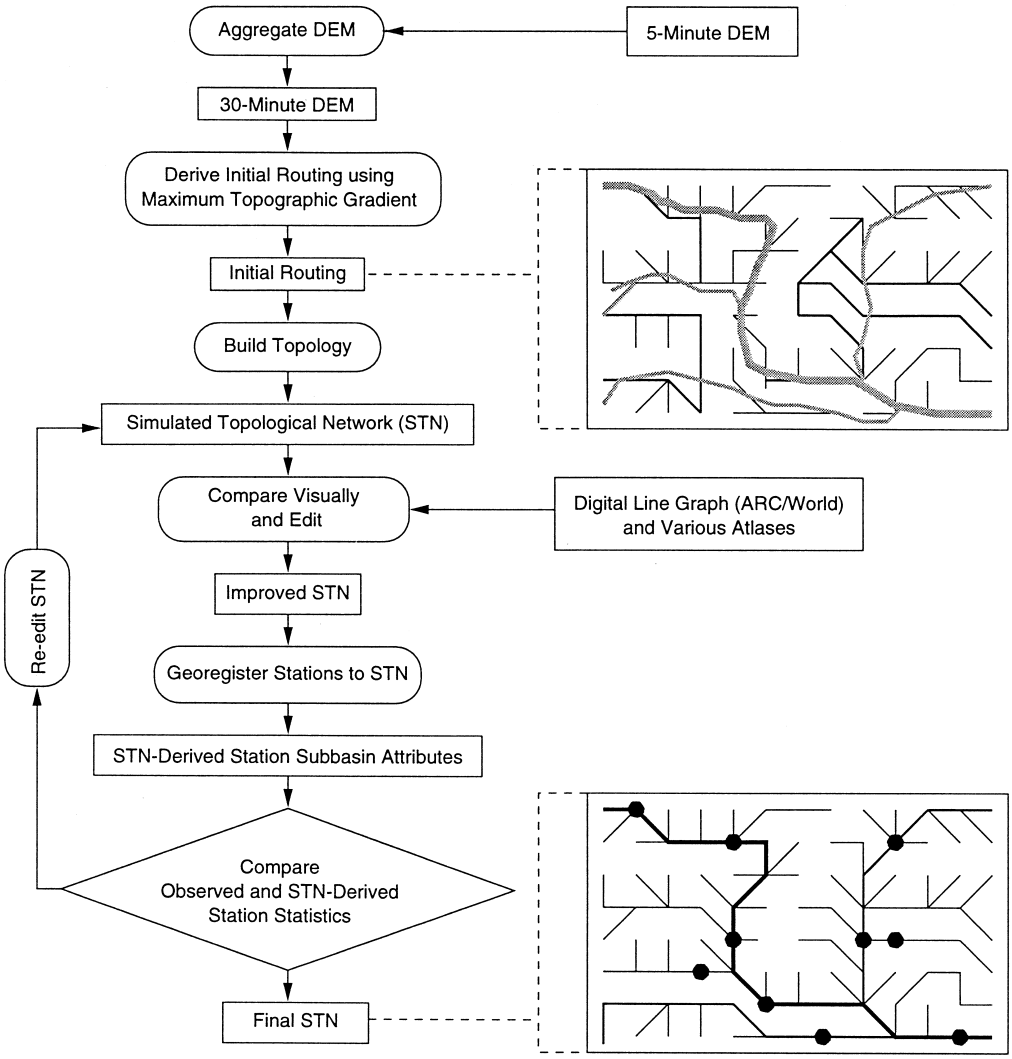


Fig. 1. Data processing stream used to construct the 30′ river networking data base. *Station* (●) refers to a river monitoring site for which independent basin and sub-basin attributes were available; *DEM* stands for digital elevation model. Arc/World 1:3M scale digital line segments (ESRI, Inc., ESRI, (1992)) were superimposed onto STN-30p flow pathways to assist in network rectification.

(Meybeck and Ragu, 1995, 1997) was from 12 to 16%, depending on whether the STN-30p length calculation was based on an area-directed search procedure or maximum segment length (see Vörösmarty et al., 2000a).

The structure of the STN-30p is determined by potential flow pathways that empty to either an ocean (exorheism) or inland receiving body (endorheism). The majority of STN-30p flow pathways (draining 87% of the land mass) connects the interior of the continents to one of four oceans (i.e. Arctic, Atlantic, Pacific, Indian) or the Mediterranean/Black Sea. An additional set of river systems empties into major land-locked receiving waters (e.g. Caspian and Aral Seas, Great Salt Lake, Lake Chad) or large topographic depressions in extremely dry regions (e.g. Takla Makan in western China). STN-30p thus defines the global network of rivers based on topographic control. The activity of the network with respect to perennial or intermittent rivers is defined

by climate. In this analysis we consider the entire set of potential rivers. The only lakes explicitly considered in STN-30p are the Caspian and Aral Seas, two endorheic receiving bodies. All other major lakes either in exorheic (e.g. American Great Lakes) or endorheic (e.g. Titicaca, Eyre) basins show potential flow paths. In regions where the drainage is poorly organized and digital elevation data problematical (e.g. eastern Chad basin) potential flow pathways are poorly defined.

GHAAS was also used to generate continental and global summaries of drainage basin attributes. The linkage of individual grid cells in STN-30p permits ordered searches to be performed on any river system and attributes to be derived for sub-basins associated with any point in the topological structure. This facility was used to determine: (a) order of interior river (line) segments, tributary mainstems, and maintems; (b) sub-basin and basin areas; (c) length of interior river segments, tributary sub-basins, and basins; (d) mean basin and sub-basin length; and, (e) elevation-related statistics.

Drainage basins and sub-basins are defined as collections of grid cells topologically connected by flow lines representing river links and segments. Stream order is based on (Strahler, 1964, 1957) and assigned to individual interior links where each link is defined as a single linear pathway connecting an adjacent pair of grid cells (Fig. 2). An ordered interior segment is defined as a single link or a group of consecutive links having the same Strahler order. At its upstream boundary a segment of order n is created by the junction of two order $n-1$ segments (for order one there are no upstream junctions); the downstream endpoint is defined by the location at which it encounters a segment of order n or greater. For each ordered interior segment we define a sub-basin above its downstream endpoint that contains all subsidiary links and associated grid cells. Each sub-basin contains a tributary mainstem. The tributary mainstem is identified using an upstream search procedure applied to the previously established topology. The upstream direction of flow for the tributary mainstem is determined by choosing the grid cell having the maximum in calculated drainage area whenever alternative upstream flow pathways are encountered. The downstream endpoint of the highest order river segment defines the order of a drainage basin as well as the position of its basin mouth. All cell, sub-basin, and basin areas were computed as a function of cell latitude. Basin and sub-basin lengths were calculated by following the tributary and basin mainstem channels as described above. All length-related computations accounted for both latitude and curvature of the Earth but not for fine-scale sinuosity (see Vörösmarty et al. 2000a).

Mean basin slope was computed from the GTOPO30 (USGS-EDC, 1996) mean elevation field, aggregated to $30'$ spatial resolution. We computed a local gradient for each STN-30p grid cell using a forward differencing scheme along the predicted flow path and then averaged the result across all cells within a drainage basin.

3. Results and discussion

In the following sections we present a set of geomorphometric attributes describing the STN-30p river networks and their associated drainage basins. We provide summaries for each of six continents and the globe, and use several individual river systems to highlight our major findings. The STN-30p river networks are shown in Fig. 3 and the drainage basins in Fig. 4.

3.1. Number of ordered river segments

The statistical distribution of river numbers shows a regular and expected pattern at individual river basin, continental, and global scales. For the land mass of the Earth, there are 33,251 unique river segments (tributary mainstems) in STN-30p (Table 1a), which, at $30'$ spatial resolution, are classified as order one through order six using the system of Strahler (Strahler, 1964, 1957). For reference, at this spatial resolution the Rhine, Chao Phraya, and Magdalena Rivers are order three at their mouths; the Mekong, Rio Grande, Orinoco, and Danube order four; the Ganges–Brahmaputra, Paraná, Zambezi, Mississippi order five; and, the Amazon and Lena order six. The most common river segment is of order one, constituting 83% of the global sum. The total number of individual river segments declines exponentially with river order. There are but two order six mainstem river segments, associated with the lower reaches of the

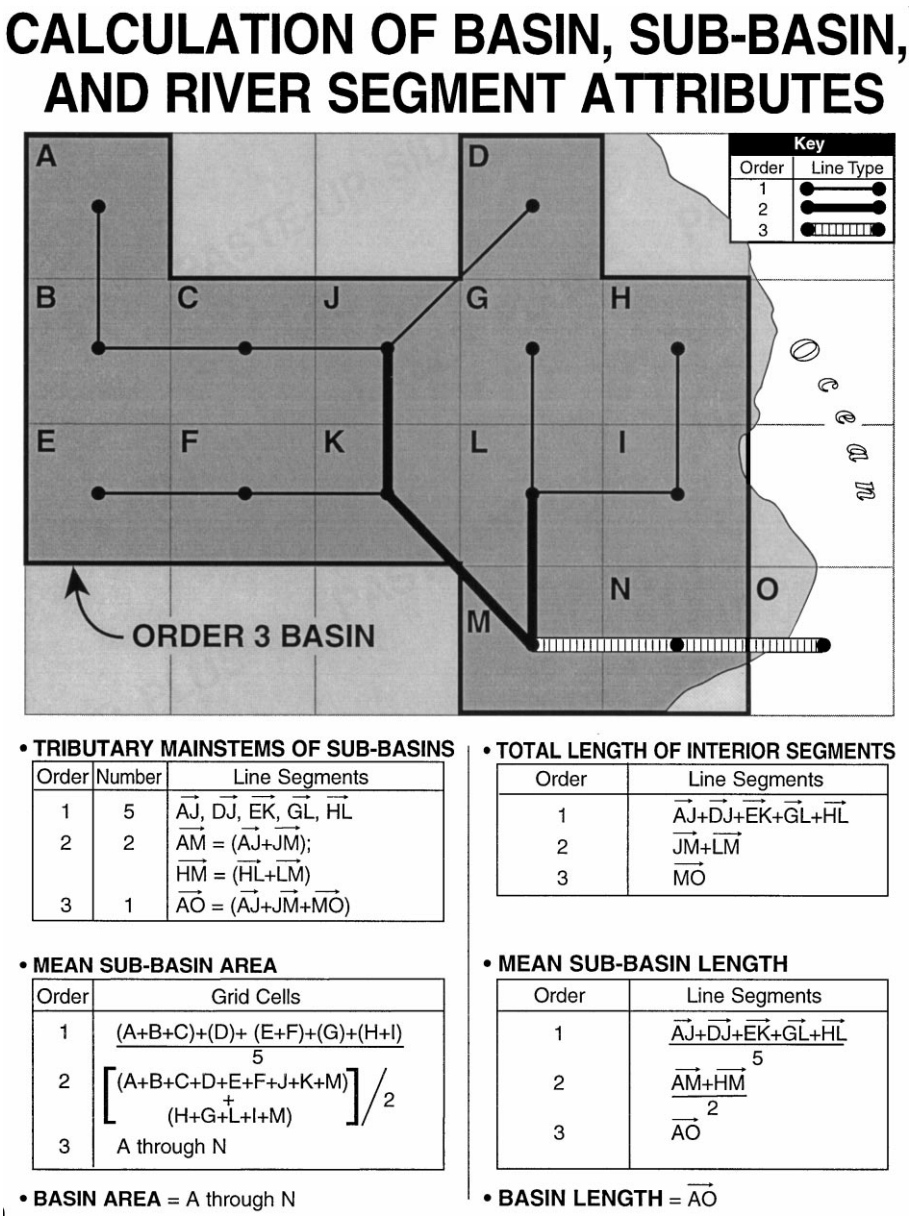


Fig. 2. Definition of river and drainage basin attributes used throughout the text and statistical summaries presented in this paper.

Amazon and Lena. The predominance of first order rivers as well as the exponential decline in segment numbers thereafter is also apparent for each of the continents.

The number of river segments (N) belonging to a single STN-30p drainage basin follows an exponential decline as a function of order. Several examples for large river systems are given in Fig. 5a. The slope of each line, R_b, approximates the bifurcation ratio (N_{w-1}/N_w) where w is a particular sub-basin river order within a basin (Horton, 1945). The relationships shown in the figure for large river basins (orders four to six) show great similarity regardless of geographic position and fall within the typical

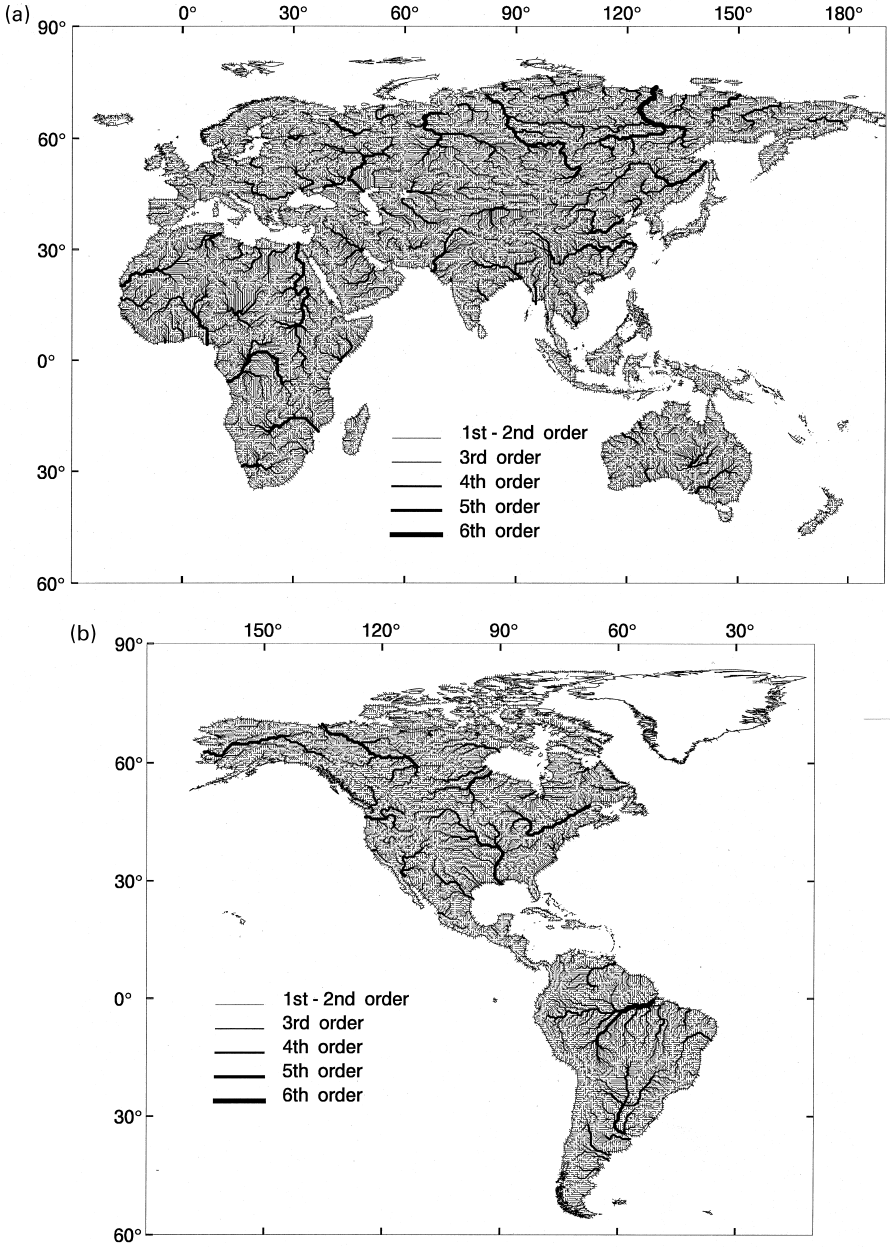


Fig. 3. STN-30p, the Simulated Topological Network for potential river systems at 30′ (longitude × latitude) spatial resolution. STN-30p provides a flow direction to 59,132 non-glacierized, land-based grid cells. Both exorheic and endorheic networks are represented in this database. *Order* refers to individual river segments. (From: Vörösmarty et al. (2000a)).

range of three to five observed at finer scales (Dingman, 1994).

The consistency of these relationships is also evident in continental and global summaries. We ran a total of 1489 individual regressions on basins of order two through six to test for the consistency suggested by Fig. 5 across a broader spectrum of rivers. For these tests, we found a global mean value

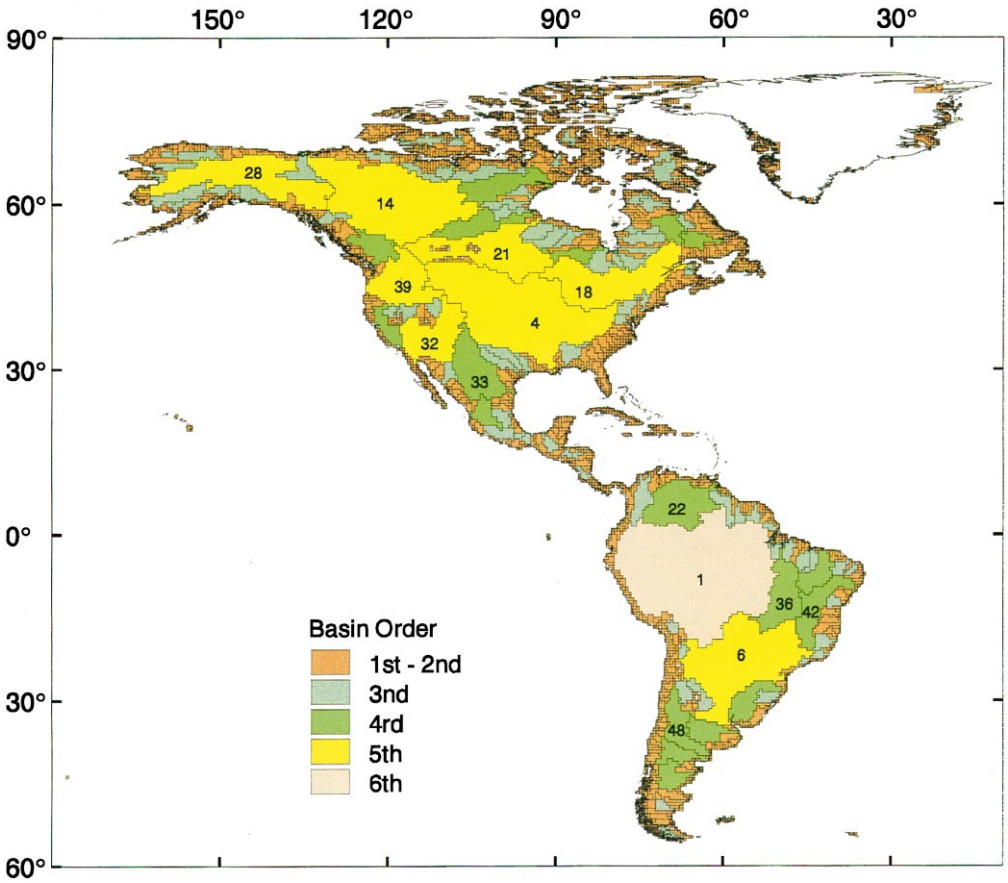


Fig. 4. STN-30p drainage basins. There are 6152 individual watersheds for the entire data set. *Order* refers to basin order defined by river segment at mouth. Basin numbers refer to individual entries in Table 3. (From: Vörösmarty et al. (2000a)).

for the bifurcation ratio of 3.48 with a mean r^2 of 0.993. If we consider the 366 basins of order three through six that better represent the geomorphic characteristics of natural basins, mean $R_b = 3.70$ and $r^2 = 0.972$. Over individual continents and the globe, bifurcation ratios tend to show a small increase with increasing basin size. For the globe, mean R_b ranges between 3.41 and 3.62 for basins of order two and three while the corresponding statistics for orders four and five are 3.90 and 3.96 (Table 1b). The two order six basins also show a high mean R_b. This pattern is repeated over most of the continents. However, the smaller landmasses of Europe and Australasia show a more ambiguous tendency.

The normalized frequency distribution of R_b is generally consistent for the larger river basins (orders three through six) as shown in Fig. 6a for the globe. The shift toward lower R_b and greater variability is apparent for orders two and three. The higher deviations associated with order two basins, in particular, suggest a potential limit of STN-30p to accurately depict network organization and resulting basin shape.

3.2. Mean sub-basin area for ordered river segments

The mean sub-basin area occupied by STN-30p river segments increases in a regular fashion and exponentially as a function of river order (Table 1c), both for individual continents and the globe. Mean sub-basin areas therefore vary widely for different orders, from a global mean of 3.2×10^3 km^2 for first order sub-basins to means of 1.5×10^6 and 4.1×10^6 km^2 for orders five and six, respectively. Across

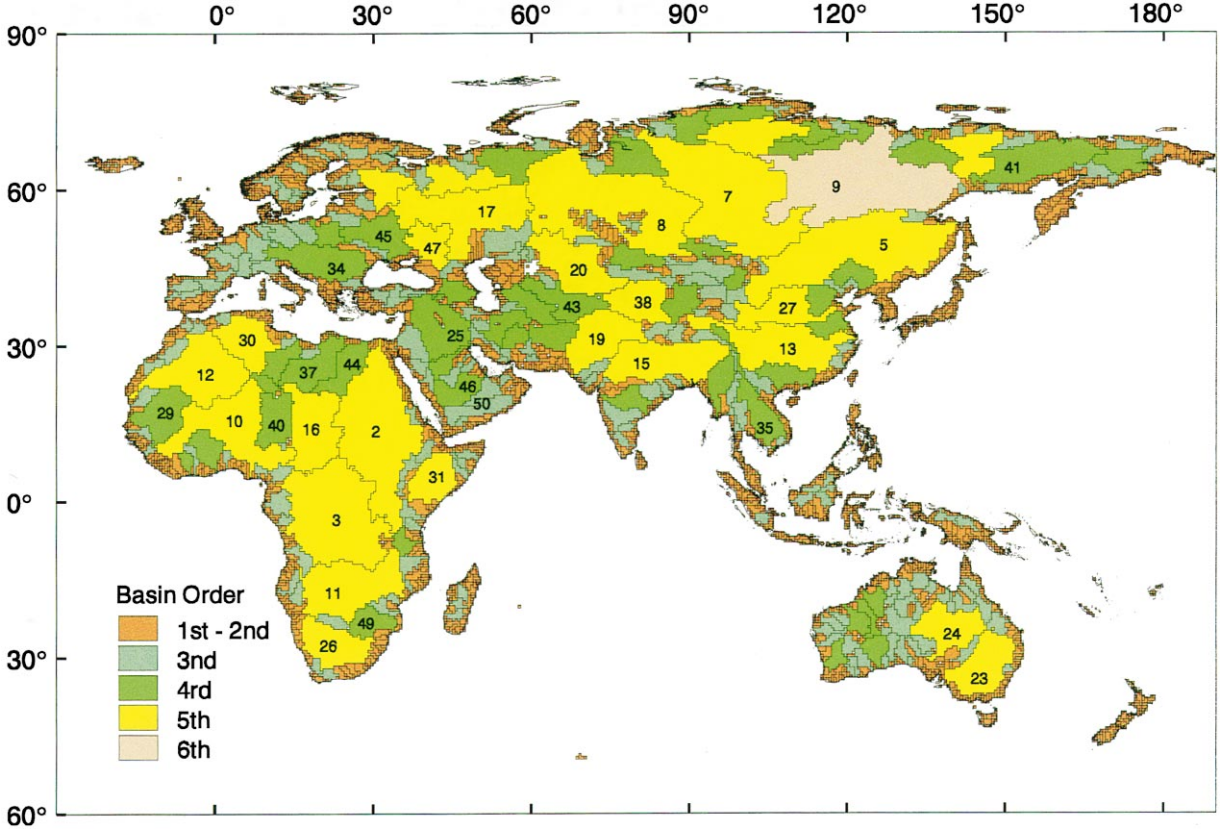


Fig. 4. (*continued*)

the continents, mean areas for sub-basins of a particular order are of roughly the same absolute magnitude for sub-basins at/or below order four. Order five has a range in continental means that varies from 0.630 to 2.330×10^6 km^2 while order six varies from 2.42 to 5.85×10^6 km^2. The maximum area is recorded for the Amazon, which is about 4% less than other estimates (Meybeck and Ragu, 1995, 1997) due to differences in including or excluding portions of the coastal zone near the mouth of the river (Fig. 4) (see Vörösmarty et al., 2000a for discussion).

Despite the broad-scale geographic differences, there is a highly stable and predictable relationship linking sub-basin area and river order for individual river systems across the globe (Fig. 5b). The slopes associated with these relationships represent the mean drainage area ratio ($R_a = A_w/A_{w-1}$) (Schumm, 1956) where A_w is sub-basin area for river order w within a larger drainage system. Fig. 5b demonstrates this pattern for large rivers in several parts of the world. The global mean value for R_a is 5.13 (mean $r^2 = 0.995$ for $n = 1489$). For the 366 order three through six basins $R_a = 4.55$ (mean $r^2 = 0.980$) and continental-scale means vary between 4.3 and 4.9 (Table 1d). The accumulation of drainage area with increasing river order in STN-30p progresses in a manner similar to that which is observed in river systems analyzed at finer spatial resolutions, and well within the range of 3 to 6 typically found in nature (Dingman, 1994).

Although composite means over large land areas behave with regularity, a greater degree of variability in R_a is apparent when we consider the entire

Table 1
Continental and global-scale summary of STN-30p tributary mainstem rivers and corresponding sub-basin attributes. Mean bifurcation ratios correspond to river systems defined by the order of each mainstem at the river mouth

		Africa	Asia	Australasia	Europe	North America	South America	GLOBAL[a]
(a) *Number of tributary mainstem rivers or sub-basins*								
Order:	1	4433	10074	1370	2825	6260	2574	27673
	2	839	1615	249	438	864	443	4456
	3	187	330	59	81	152	97	906
	4	41	65	10	14	26	20	176
	5	9	13	2	4	7	3	38
	6	–	1	–	–	–	1	2
Total		5509	12098	1690	3362	7309	3136	33251
(b) *Mean bifurcation ratio* (R_b) [b,c]								
Order:	2	3.15	3.37	3.38	3.70	3.48	3.36	3.41
	3	3.41	3.67	3.49	3.96	3.70	3.24	3.62
	4	3.82	4.00	3.15	4.44	3.75	3.94	3.90
	5	3.96	4.06	3.55	3.45	4.17	4.40	3.96
	6	–	3.78	–	–	–	3.82	3.80
Mean (Orders 3–6)		3.57	3.78	3.45	3.97	3.75	3.49	3.70
(c) *Mean sub-basin area* (10^3 km^2)								
Order:	1	4.4	3.0	3.8	2.5	2.5	4.6	3.2
	2	22.6	16.3	19.5	14.0	14.9	24.5	18.0
	3	98.3	75.0	79.5	62.4	73.8	105.2	82.0
	4	399	354	278	273	305	554	369
	5	1970	1350	1000	634	1370	2330	1490
	6	–	2420	–	–	–	5850	4140
(d) *Mean drainage area ratio* (R_a)[d]								
Order:	2	4.90	5.19	5.05	5.77	5.54	5.42	5.32
	3	4.28	4.59	4.43	5.00	4.75	4.15	4.57
	4	4.44	4.69	3.74	5.13	4.45	4.62	4.57
	5	4.42	4.56	4.06	3.88	4.55	4.97	4.43
	6	–	4.11	–	–	–	4.18	4.15
Mean (Orders 3–6)		4.33	4.61	4.32	4.91	4.69	4.31	4.55

Table 1 (*continued*)

		Africa	Asia	Australasia	Europe	North America	South America	GLOBAL[a]
(e) *Mean sub-basin length* (km)[e]								
Order:	1	91	76	84	72	71	92	79
	2	281	238	245	222	228	304	249
	3	637	563	565	507	551	704	586
	4	1240	1340	1110	1030	1220	1630	1300
	5	2860	2860	1410	1640	2770	2930	2645
	6	–	4387	–	–	–	4330	4360
(f) *Mean length ratio* (R_l)[f]								
Order:	2	3.06	3.11	3.02	3.26	3.26	3.27	3.17
	3	2.53	2.57	2.55	2.72	2.68	2.52	2.60
	4	2.38	2.48	2.32	2.64	2.51	2.59	2.48
	5	2.28	2.41	2.02	2.13	2.42	2.34	2.32
	6	–	2.30	–	–	–	2.12	2.21
Mean (Orders 3–6)		2.47	2.53	2.49	2.65	2.63	2.52	2.55

[a] Global totals incorporate the statistics for several small islands.

[b] R_b is the bifurcation ratio across a series of stream orders in an individual basin. We applied least squares regression to approximate R_b by the line: $\log_{10}N_w = b_{0b} - log_{10}R_b w$, where N_w is the number of streams, b_{0b} is the vertical axis intercept, and w is a given order. Entries in this table represent the mean of all computed R_b values for each continent or the globe. All relationships used to compute mean bifurcation ratios yielded r^2 greater than 0.891 and were significant at the $p < 0.0001$ level. Mean r^2 was 0.993 on $n = 1489$ regressions.

[c] Number of basins given in Table 2.

[d] R_a is the drainage area ratio across a series of stream orders in an individual basin. We applied least squares regression to approximate R_a by the line: $\log_{10}A_w = b_{0a} + \log_{10}R_a w$, where A_w is the number of streams, b_{0a} is the vertical axis intercept and w is a given order. Entries in this table represent the mean of all computed R_a values for each continent or the globe. All relationships used to compute mean bifurcation ratios yielded r^2 greater than 0.893 and were significant at the $p < 0.0001$ level. Mean r^2 was 0.995 on $n = 1489$ regressions.

[e] Based on mainstem length determined by area-directed upstream search.

[f] R_l is the length ratio across a series of stream orders in an individual basin. We applied least squares regression to approximate R_l by the line: $\log_{10}L_w = b_{0l} + \log_{10}R_l w$, where L_w is the mean length of streams, b_{01} is the vertical axis intercept and w is a given order. Entries in this table represent the mean of all computed R_l values for each continent or the globe. All relationships used to compute mean bifurcation ratios yielded r^2 greater than 0.729 and were significant at the $p < 0.0001$ level. Mean r^2 was 0.991 on $n = 1489$ regressions.

population of individual basins for a particular order. Fig. 6b shows the normalized frequency distributions of basin-wide calculated R_a. Orders four through six show a convergent pattern. The means for order three and two are comparable to the higher order rivers (Table 1d). However, the numerical dispersion increases for orders three and two, suggesting again a possible limit to the STN-30p in accurately depicting the relatively small basins.

3.3. Mean sub-basin length for ordered river segments

Mean accumulated segment length increases in a regular fashion for internal sub-basins from order one through order six (Table 1e). At the global scale order one rivers have a mean of 80 km. Thereafter, there is an approximate doubling of mean length for each subsequent sub-basin order: 250 km for order two followed by 600, 1300, 2700 and 4400 km for the remaining orders. Up to order three, the absolute range in mean sub-basin lengths varies little for each order over different continents. There is generally more variability at the higher sub-basin orders. In particular order five continental means vary from a low of 1400 km to a high of 2900 km. Both order six rivers (Amazon and Lena) have lengths of approximately 4350 km. As discussed earlier (see Section 2) fine-scale sinuosity is not taken into account at 30′ spatial resolution and lengths in STN-30p differ from those published elsewhere for some major world rivers.

Fig. 5c shows several examples of the relationship between accumulated river length and sub-basin order. Across sequential orders, the mean length of

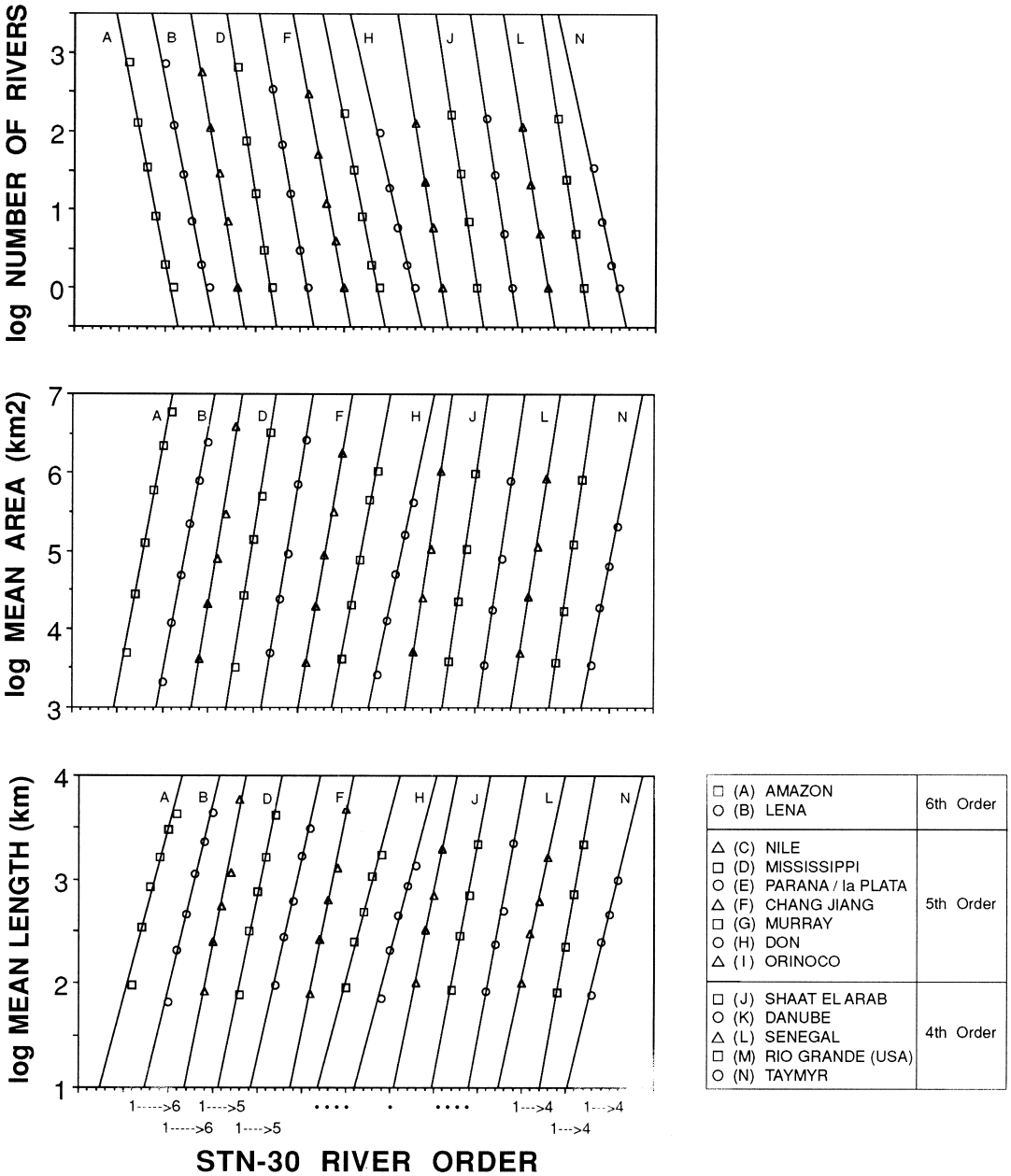


Fig. 5. Examples of large STN-30p river systems showing the relationship between interior tributary mainstems classified by order (according to Strahler, 1964) and (top) the number of individual river segments, (middle) mean area of each contributing sub-basin, and (bottom) mean accumulated length. Basins of order four through six at river mouth are shown. Reading from left to right, the horizontal axis is a repeating cycle of subsidiary STN-30p stream orders for each basin, from order = 1 to maximum order at mouth (see Horton, 1945). The bifurcation ratio can be approximated by the slope of each fitted line through $R_b = 10^{-\text{slope}}$. The drainage area ratio (R_a) and length ratio (R_l) share a similar mathematical form but with positive exponents. Continental and global-scale summaries of R_b, R_a, and R_l are given in Table 2.

Table 2
Continental and global-scale summary of STN-30p river basin attributes. Statistics correspond to river systems defined by the order of each mainstem at river mouth

		Africa	Asia	Australasia	Europe	North America	South America	GLOBAL[a]
(a) *Aggregate basin area* (10^6 km^2)								
All orders		30.1	44.4	7.8	10.1	22.4	17.9	133.1
(b) *Number of basins*								
Order:	1	326	1470	209	621	1591	331	4663
	2	152	355	74	138	302	94	1123
	3	39	86	32	33	54	21	265
	4	10	27	5	5	9	9	65
	5	9	11	2	4	7	1	34
	6	–	1	–	–	–	1	2
All orders		536	1950	322	801	1963	457	6152
(c) *Mean basin area* (10^3 km^2)								
Order:	1	5.0	3.1	4.5	2.4	2.1	4.4	3.0
	2	20.2	15.6	17.9	13.9	12.8	21.7	16.0
	3	81.8	78.3	79.7	64.7	58.3	81.1	73.4
	4	444	336	191	400	270	464	355
	5	1970	1460	1000	634	1370	2660	1490
	6	–	2420	–	–	–	5850	4140
(d) *Mean basin length* (km)[b]								
Order:	1	107	80.2	98.4	75.7	71.5	95.0	80.2
	2	260	227	220	221	214	278	231
	3	584	553	554	531	512	580	549
	4	1316	1290	984	1372	1188	1535	1296
	5	2846	2951	1378	1630	2767	3072	2641
	6	–	4387	–	–	–	4327	4357
(e) *Mean basin shape index* (S_b)[c]								
Order:	1	1.48	1.49	1.43	1.60	1.68	1.41	1.56
	2	1.83	1.89	1.64	1.92	1.95	1.89	1.88
	3	2.06	2.04	2.01	2.16	2.15	2.09	2.08
	4	2.09	2.26	2.27	2.19	2.38	2.35	2.25
	5	2.01	2.48	1.37	2.12	2.42	1.88	2.22
	6	–	2.82	–	–	–	1.79	2.30
Mean (Orders 3–6)		2.06	2.13	2.01	2.16	2.21	2.15	2.12
(f) *Mean Schumm elongation ratio* (E_r)[d]								
Order:	1	0.82	0.84	0.86	0.79	0.81	0.85	0.82
	2	0.65	0.64	0.73	0.64	0.65	0.63	0.65
	3	0.58	0.59	0.58	0.54	0.55	0.56	0.57
	4	0.55	0.56	0.51	0.53	0.49	0.49	0.53
	5	0.59	0.51	0.88	0.54	0.48	0.60	0.55
	6	–	0.40	–	–	–	0.63	0.52
Mean (Orders 3–6)		0.58	0.57	0.59	0.54	0.54	0.54	0.56
(g) *Mean basin slope* (m/km)								
Order:	1	3.40	3.33	3.00	2.94	2.87	5.83	3.20
	2	3.77	3.52	1.70	2.47	2.63	6.00	3.28
	3	2.61	3.11	0.93	2.03	2.28	3.27	2.49
	4	1.05	2.66	0.66	2.21	2.10	2.23	2.17
	5	1.34	2.72	0.80	0.52	2.20	1.59	2.19
	6	–	1.83	–	–	–	1.73	1.97
Global mean		1.75	2.81	1.27	1.84	2.36	2.74	2.38

[a] Global totals incorporate the statistics for several small islands.
[b] Based on mainstem length determined by area-directed upstream search.
[c] Computed as the mean of all $S_b = L/A^{0.5}$ where L is maximum length (km) and A is basin area (km^2).
[d] Computed as the mean of all individual basin ratios where E_r = diameter of circle of same area/maximum basin length (Schumm, 1956).

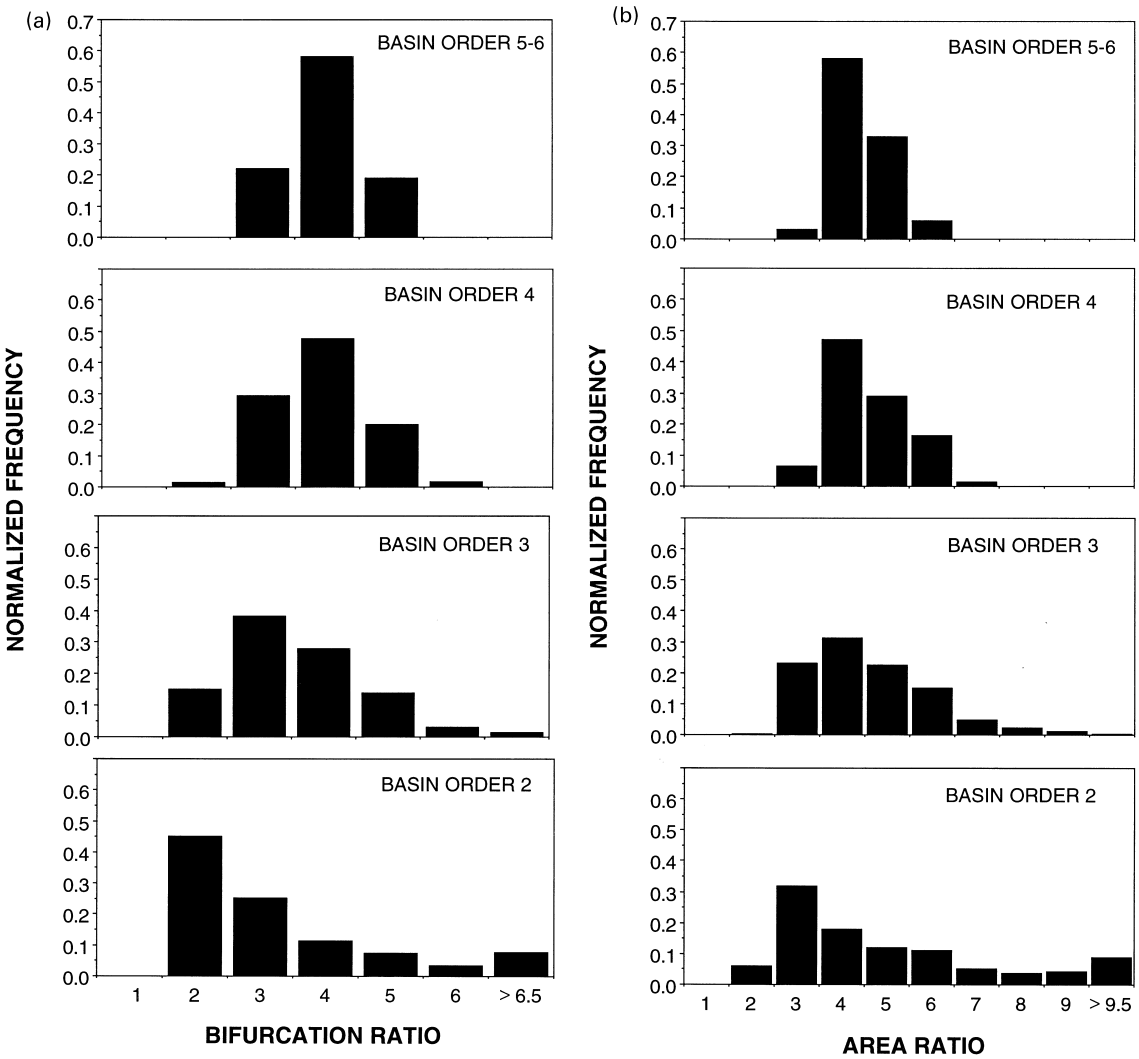


Fig. 6. Normalized frequency distributions of the (a) mean bifurcation ratio R_b, (b) mean area ratio R_a, and (c) mean length ratio R_l. These distributions were derived by analyzing all river systems of the STN-30p, classified according to basin order at mouth.

sub-basin increases in exponential fashion and with great regularity. A length ratio can be computed as $R_l = L_w/L_{w-1}$ (Horton, 1945), where L_w is sub-basin length for river order w within a larger drainage system. The overall mean value for R_l is 3.02 (mean $r^2 = 0.991$ for $n = 1489$. The 366 basins of order three through six have a lower mean, $R_l = 2.55$ ($r^2 = 0.963$) (Table 1f). For the globe, there is a small but progressive decrease in this ratio with increasing basin order, from 3.2 to 2.2 for order two through order six, respectively, although orders three through six show most consistency. Similar trends are noted for each of the continents. The variation in mean R_l for particular orders across the continents is small; the overall range in mean R_l for all orders across individual continents is from 2.90 to 3.15, while for basin orders three through six it is 2.47 to 2.65. STN-30p basins are similar to river systems analyzed elsewhere, which typically show a range in R_l of 1.5 to 3.5 (Dingman, 1994). As was the case for R_b and R_a, these expected values show a numerical dispersion across individual drainage basins. The variability is

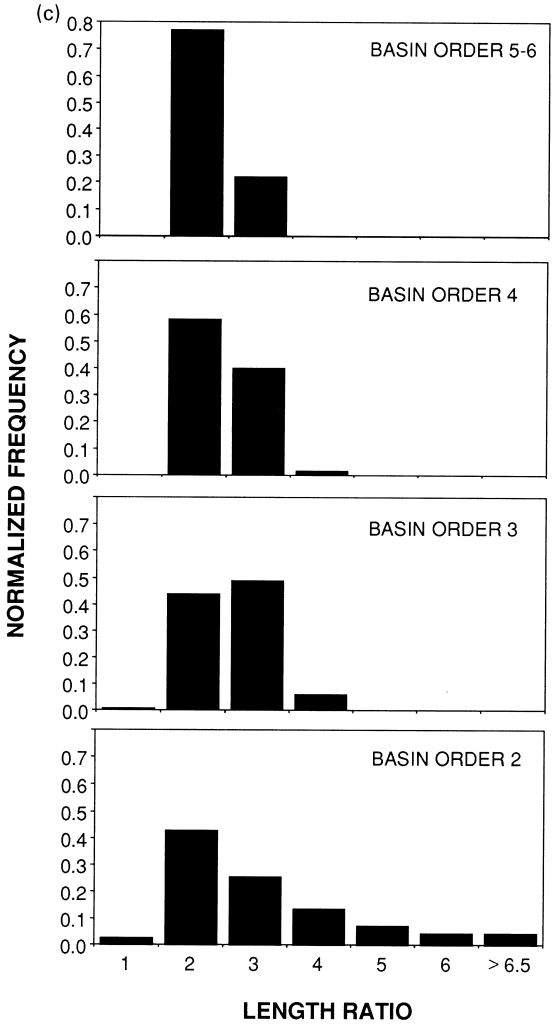


Fig. 6. (*continued*)

stable for basin orders three through six but increases for order two (Fig. 6c). The predictability of length-related attributes for small STN-30p basins is therefore low.

3.4. Number of STN-30p drainage basins

There are 6152 individual watersheds represented by the STN-30p for the 133.1×10^6 km^2 of non-glacierized global land area (Table 2a). As expected from their relative size, North America and Asia show the largest total number of STN-30p watersheds while Australasia shows the smallest. Africa, Europe, and South America are intermediate. Starting at order one, the total number of drainage basins declines rapidly as a function of river order at mouth (Table 2b). There are nearly 5800 STN-30p basins of first and second order, more than 90% of all simulated watersheds. There are only 366 basins of order three or greater with the majority of these remaining systems classified as order three. Larger river systems are few. There are only 65 order four and 34 order five basins. The Amazon and Lena River systems are the only order six basins in the STN-30p.

3.5. Mean drainage basin area

Not unexpectedly, there is a progressive increase in mean basin size accompanying increasing river order (Table 2c). Mean area spans several orders of magnitude, from approximately 10^3 km^2 for first order basins to $>10^6$ km^2 for orders five and six. Across the continents, mean basin area for order three rivers is relatively consistent, varying by 20% or less around the global mean. Larger proportional differences are evident for the remaining orders.

3.6. Mean basin (mainstem) length

For the globe, the mean length of an STN-30p basin (i.e. length of mainstem) increases across orders from less than 100 to more than 4300 km (Table 2d). Each continent is more or less surrounded by a fringe of small to medium-sized basins at the land–ocean (or land–internal receiving body) boundary (Fig. 4). Large basins extend well into the continents and drain runoff sometimes thousands of kilometers from basin outlets. For the 50 largest river systems (ranked by area) that drain 53% of the land (Vörösmarty et al., 2000a) all have mainstem lengths in excess of 1000 km and a mean of 2590 km. The remaining 6102 river basins show a mean of 137 km and only about 40 have lengths exceeding 1000 km. The continents show consistency in mainstem lengths associated with basin orders one through three. Ranges in means for orders one, two, and three are approximately 70–110, 215–280, and 510–580 km, respectively. The mean lengths at the continental scale for order five and six basins exceed 1000 km. Globally and for orders one through six, a total of 3.24×10^6 km of length is represented by STN-30p river segments.

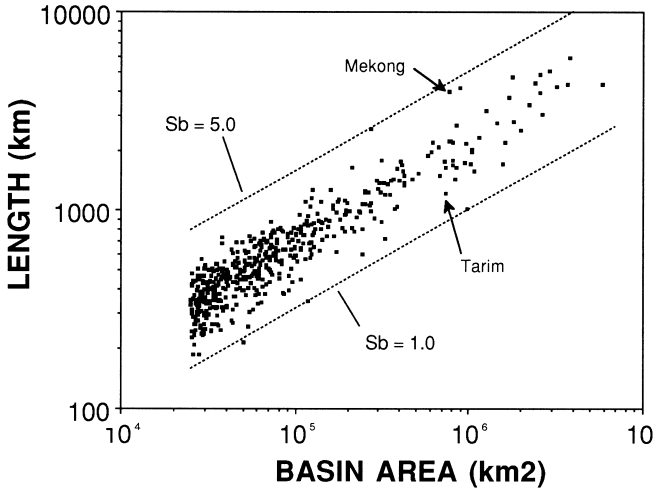


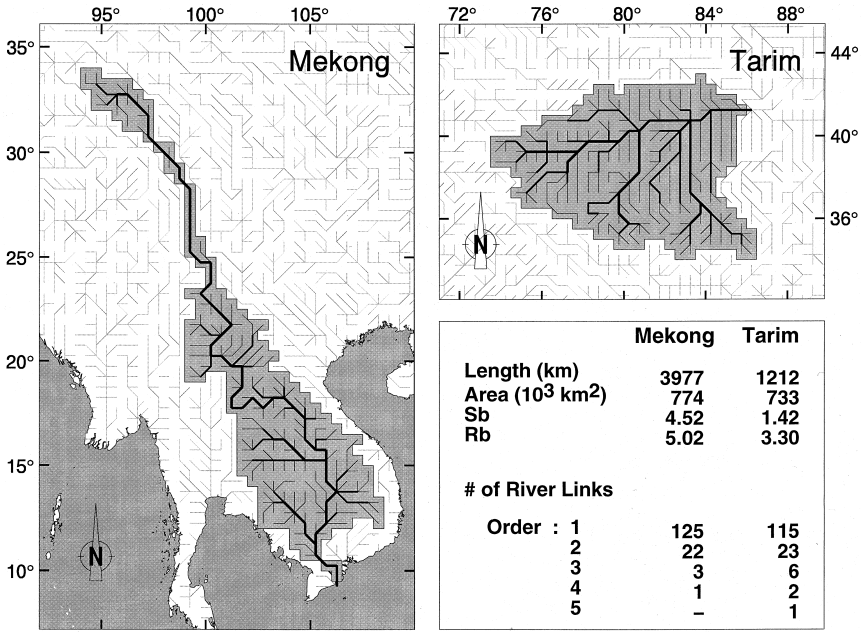
Fig. 7. The relation between basin length and basin area in STN-30p. All basins 25,000 km^2 or greater were considered to have reliable area and length estimates. S_b is the shape index as defined in the text. The world average relationship is $\log L = 0.523 \log A + 0.2141$ ($r^2 = 0.865$, $\alpha < 0.001$, $n = 522$). Details for the Mekong and Tarim river systems are given in Fig. 8.

3.7. Length/area relations and basin shape

Mainstem river length is generally well-correlated with drainage basin area (Fig. 7). Leopold et al. (1964) showed this relationship for 34 large rivers of the world, ranging from about 10^4 to $>10^6$ km^2. An equation to describe this relation is $L = 1.51A^{0.560}$ where L is maximum river length in km and A is area in km^2. This relationship is nearly identical for STN-30p basins exceeding 25×10^3 km^2 (i.e. those with the highest correspondence to independently published area estimates) and is $L = 1.64A^{0.523}$ ($r^2 = 0.87$; $n = 555$; $p < 0.01$). There is a systematic underestimate by STN-30p since it does not account for fine-scale sinuosity. The predicted length from the STN-30p equation expressed as a fraction of that predicted by the Leopold et al. relationship varies progressively from 0.75 to 0.63 for basins with areas between 2.5×10^4 and 2.5×10^5 km^2, respectively. This corresponds well to the underestimates noted in our work on network verification (Vörösmarty et al., 2000a). Runoff or constituent routing algorithms would thus have to assign correction factors to accomodate the bias.

The variability shown in Fig. 7 reflects differences in the shape of individual drainage basins. We computed a simple shape index ($S_b = L/A^{0.5}$) for each STN-30p basin where L is mainstem basin length (km) and A is drainage area (km^2) at river mouth. High values of S_b indicate an elongated river network while low values represent a more rounded configuration. The shape index for small STN-30p rivers, such as for order one and two, are not reliable because of difficulties in reproducing basin shape from rectangular grids. For basin orders three through six the mean value of S_b is remarkably stable across all continents, varying between 2.01 and 2.21, and having a global value of 2.12 (Table 2e). For individual basins exceeding 25,000 km^2, the full range in S_b is from 1.0 to 5.0 (Fig. 7).

The Mekong and Tarim River basins, which have significantly different network configurations, exemplify this contrast (Fig. 8). Although both basins have drainage areas of about 750×10^3 km^2, the Mekong is more than three times longer than the Tarim, with a length of nearly 4000 km versus 1200 km and a corresponding S_b value of 4.52 versus 1.42. Both rivers have around 150 individual ordered river segments. However, since a high degree of elongation infers that the accumulation of larger order segments will be inefficient (because several lower order tributaries discharge directly into a higher order mainstem with no corresponding increase in order), the Mekong shows a bifurcation ratio of 5.02 whereas the Tarim gives 3.30. From this standpoint, the



	Mekong	Tarim
Length (km)	3977	1212
Area (10^3 km^2)	774	733
Sb	4.52	1.42
Rb	5.02	3.30
# of River Links		
Order : 1	125	115
2	22	23
3	3	6
4	1	2
5	–	1

Fig. 8. Some characteristics of the Mekong and Tarim Rivers, representing contrasting drainage basin shapes. The map displays potential river systems and their associated basins at 30′ spatial resolution.

Tarim accumulates order more efficiently and the result is a fifth order river system at its mouth. The Mekong is relatively less efficient and is fourth order at its mouth. There are several other large basins like the Mekong that drain mountainous regions and that show an apparent tectonic control of their river networking structure. These include the Upper Indus, Brahmaputra, Salween, Irrawaddy, Ucayali and Marañón Rivers.

We also computed the elongation ratio of Schumm (1956) (E_r = diameter of circle of same area/maximum basin length). This ratio measures how closely a drainage basin resembles a circle, with higher values associated with increased roundness. Schumm predicted E_r to vary between 0.6 and 1.0 over a wide variety of basins in different climates and geology. The global mean and median values for all STN-30p basins order three through six (0.56 and 0.54; Table 2f) fall slightly below the range, although 30% of all values fall within it. The continental means for E_r vary little between 0.54 and 0.59. STN-30p therefore shows basins that are generally more elongated than what was predicted by Schumm.

3.8. Mean basin slope

Mean basin slope along river courses varies across stream order at both continental and global scales and in response to the topographic environment of individual drainage basins. The overall mean is 2.38 m km^{-1}. Globally, mean slopes vary over all basin orders from 2.0 to 3.3 m km^{-1}, with highest values for order one to two followed by a decrease from the maximum for orders three through six (Table 2g). Although there are many low-order, low relief coastal plain rivers (Fig. 4), there are also numerous small rivers that occur in mountainous regions that discharge directly into the ocean. Collectively, such high relief coastal basins raise the overall mean for these lower orders. Beyond order two, large river systems are organized over broad spatial domains that include not only high elevation zones but also extensive lowlands that tend to diminish the overall mean value.

Mean slope varies across continents from 1.30 to 2.81 m km^{-1} (Table 2g). Australasia represents the continent with lowest slope due to the general absence of high elevation mountains. Asia and South America,

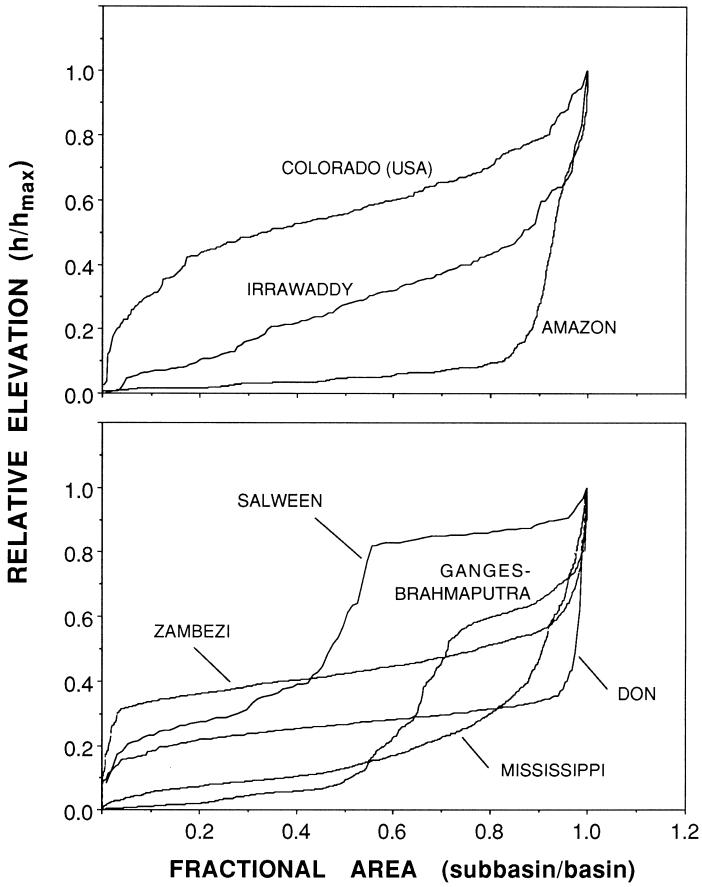


Fig. 9. Hypsometric curves for large STN-30p rivers normalized to maximum basin altitude and basin area. Examples of basins showing varying stages of maturity from active mountain uplift (Salween) to stability (Amazon, Don, Mississippi). Plateau incision is evident from convex profiles (Colorado, Zambezi). The Ganges–Brahmaputra shows a composite of an active uplift (Brahmaputra) and stable (Ganges) profile. (h_{max} is the maximum elevation recorded in each basin).

on the other hand, show the greatest values. The high mean value for Asia is the result of broad mountain belts that extend over much of the southern and eastern portion of the continent (i.e. Hindu Kush, Himalayas, Tien Shan, and the various mountain ranges of eastern Siberia). For South America, many low-order basins discharge westward into the Pacific Ocean only a short distance from source areas located in the Andes Mountains, thus boosting overall mean slope.

Several patterns of average basin elevation profiles along river courses can be described (Fig. 9). These are based on the relationship between two normalized variables, the relative elevation (elevation at given location/maximum elevation) and fractional area drained (sub-basin area at given location/total basin area) (Strahler, 1957). Both the Don and Amazon Rivers illustrate common concave profiles. Although their maximum elevations are very different, in both basins only a small portion of total area shows a relatively high altitude. The vast majority of each basin is lowland. Ninety percent of each basin has an altitude lower than 30% (Don) and 20% (Amazon) of the maximum altitude; 50% of each basin's area has less than 25% (Don) and 5% (Amazon) of the maximum elevation. The Colorado basin profile is quite different and convex. Due to the Colorado Plateau about 50% of the basin is still at an altitude higher than 50% of maximum elevation. The difference between the convex and concave types is explained here by the occurrence of the deeply incised Colorado

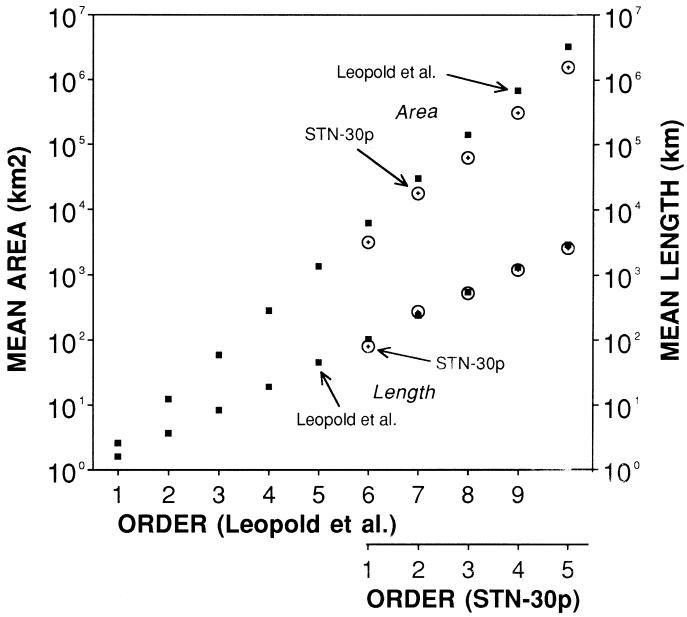


Fig. 10. Mean area and stream length as a function of order. Data from Leopold et al. (1964) represents a 1:62,500 scale map interpretation for the conterminous United States. STN-30p outputs are for basins in North America ($n = 779$) with the exclusion of the St. Lawrence and Arctic region rivers.

canyon (up to 1600 m). Such convex profiles are relatively infrequent among large basins. The Salween River in southeast Asia also displays a high convexity corresponding to major relief incision. In its upper reaches about 40% of the basin area is still at an altitude higher than 80% of the maximum elevation. The Irrawaddy has an intermediate profile. The Zambezi profile is, like the Amazon or Don, very much concave except near the coastline reflecting the sudden drop in elevation and increase in river slope downstream of the Cabora Bassa dam in Mozambique. The Ganges–Brahmaputra system presents a very unusual pattern, which is a composite of the Brahmaputra, which has a profile much like that of the Salween, and the Ganges, which is similar to the Amazon.

These hypsometric relationships bear important consequences to constituent transport. For suspended sediment, for example, concave profiles are typically linked with sediment mobilization occurring first in the periphery of a basin followed by effective retention inside depositional areas, which can include extensive floodplains (Walling, 1983). These systems bear the imprint of a long-term history of erosion and of deposition in stable basins. In contrast, convex systems contain more potential headwater landscape that is vulnerable to erosion and relatively less lowland available for sediment retention. A convex basin is relatively efficient at mobilizing sediment from headwater areas and transporting it through river networks to the basin mouth. Convex profiles in mid-basins characterize deep plateau incision corresponding to channel erosion as in the Colorado (Fig. 9).

3.9. STN-30p in relation to finer-scale river networks

The structure of STN-30p river systems appears consistent with those of rivers analyzed at finer spatial scales as demonstrated by the numerical similarity of several geomorphometric indices including bifurcation, drainage area, basin length, and Schumm elongation ratios. We find this remarkable since much of the detail inherent in more localized studies of the geomorphology of river systems has been subsumed within the elemental 30′ grid cell of the STN-30p. A study of drainage systems in the conterminous United States (Leopold et al., 1964) based on 1:62,500 maps helps support this contention (Fig. 10). From

comparison of the plots relating mean basin area and length as a function of order we see that STN-30p basin order 1.0 is equivalent to an order 5.5 basin for area and 6.0 for length derived from the 1:62,500 maps. Further, STN-30p behaves in virtually the same way as the rivers derived from the finer-scale maps.

This consistency has important ramifications since it suggests that established methods of river network analysis can be employed over much broader spatial domains. Further it confirms that the STN-30p has preserved the inherent organization of river systems, albeit at a coarser spatial resolution. The challenge in using such a data set for global change studies will rest on our ability to progressively scale riverine fluxes of water and constituents, obtained typically at the local scale, to the global domain.

3.10. Summary for major river basins

Because large river systems drain a significant portion of individual continent and global land masses we provide a listing of key drainage basin attributes (Table 3) for the 50 largest river basins (ranked by area) that together drain approximately 50% of the Earth's land mass (Vörösmarty et al., 2000a). These attributes include basic geographical and physical data discussed above including its endorheic/exorheic status, order at mouth, drainage area, basin length, mean slope, mouth/maximum/mean elevation, and alternate shape factors.

4. Conclusions

We have analyzed the spatial organization of the global land mass using a simulated topological network (STN-30p) representing potential flow pathways across the entire non-glacierized surface of the Earth at 30′ (longitude × latitude) spatial resolution. We derived from STN-30p a set of geomorphometric statistics on river segments defining sub-basins, complete drainage basins, individual continents, ocean basins, and the globe.

From both our study of individual stream segments reported here and basin-scale analysis (Vörösmarty et al., 2000a) we have highest confidence in the depiction by STN-30p of large river systems (>25,000 km^2). There are 522 such basins draining a potential area of 109×10^6 km^2 or 82% of the land mass of the Earth. We also believe that composite statistics of river systems are generally sound across all size classes. The numerical characteristics of these STN-30p river systems behave similar to those of drainage basins analyzed at finer spatial resolution and suggest the fidelity of the overall database in global change studies. We have less confidence in the representation of the 5630 individual drainage basins smaller than 25,000 km^2. For such basins, derived statistics such as for length, shape, or relief should therefore be viewed with due caution. For global climate and biogeochemical studies, a composite of the 30′ resolution and finer spatial resolutions appears to be necessary.

Many of the limitations inherent within a 30′ topology in principle can be substantially reduced with the advent of high-resolution 1-km global elevation models and river networks (USGS-EDC, 1998, 1996). Nonetheless, we see distinct advantages to using the STN-30p operating at 30′ spatial resolution. First, it has been geographically co-registered to both the UNESCO/RivDIS (Vörösmarty et al., 1998a, 1996b) and GEMS/GLORI (Meybeck and Ragu, 1997, 1995) data banks to facilitate calibration and validation of drainage basin models. Second, its computational burden and disk storage requirements are modest in comparison to a 1-km topology, especially when considering continental-to-global scale simulations. Importantly, the probability of undiscovered errors in network configuration is much higher at the fine scale.

STN-30p was developed expressly to support a range of Earth systems studies requiring digital river basin information at or close to 30′ spatial resolution. Spatially-distributed runoff fields from macro-scale water balance models (e.g. Arnell, 1995; Vörösmarty et al., 1998b, 1996a, 1991, 1989; Mintz and Serafini, 1989) and/or climate simulations (e.g. Kite et al., 1994; Sausen and Dümenil, 1994; Miller and Russell, 1992), when linked to topologically-organized river systems, provide a mechanism to validate model outputs by using well-established, station-based records of river discharge. The organization of water budgets through drainage basins and river systems as well can provide support to broad-scale assessments of global climate change on terrestrial water systems (Arnell et al., 1996; Kaczmarek et al., 1996). Assessment of the disturbance of the terrestrial water cycle

Table 3
Characteristics of the 50 largest potential river systems of the world ranked by area in STN-30p. These rivers collectively drain 53% of the continental land mass (Vörösmarty et al., 2000a). Statistics are defined and described in the text and refer to both rheic (discharging) and arheic (non-discharging) portions of each drainage system

Rank	Name	Continent[a]	Order	Area (10^6 km^2)	Length (km)	Basin shape (S_b)	Elongation ratio (Schumm)	Mean slope (m km^{-1})	Mean elev. (m)	Max. elev. (m)	Elev. of mouth (m)
1	Amazon	SAM	6	5.854	4327	1.79	0.63	1.66	430	6600	0
2	Nile	AFR	5	3.826	5909	3.02	0.37	1.45	690	4660	0
3	Zaire	AFR	5	3.699	4339	2.26	0.50	1.11	740	4420	0
4	Mississippi	NAM	5	3.203	4185	2.34	0.48	1.66	680	4330	0
5	Amur	ASIA	5	2.903	5061	2.96	0.38	1.80	750	5040	0
6	Parana	SAM	5	2.661	2748	1.88	0.60	1.59	560	6310	0
7	Yenisei	ASIA	5	2.582	4803	3.00	0.38	1.94	670	3500	0
8	Ob	ASIA	5	2.570	3977	2.46	0.46	1.28	270	4280	0
9	Lena	ASIA	6	2.418	4387	2.82	0.40	1.83	560	2830	0
10	Niger	AFR	5	2.240	3401	2.27	0.50	0.94	410	2980	0
11	Zambezi	AFR	5	1.989	2541	1.80	0.63	1.61	1050	2970	0
12	Tamanrasett[b]	AFR	5	1.819	2777	2.07	0.54	0.83	450	3740	0
13	Chang Jiang	ASIA	5	1.794	4734	3.53	0.32	3.27	1660	7210	0
14	Mackenzie	NAM	5	1.713	3679	2.84	0.40	2.23	590	3350	0
15	Ganges–Brahmaputra	ASIA	5	1.628	2221	1.74	0.65	6.00	1620	9720[c]	0
16	Chari[d]	AFR	5	1.572	1733	1.38	0.82	1.10	510	3400	260
17	Volga[d]	EUR	5	1.463	2785	2.30	0.49	0.52	170	1600	− 40
18	St. Lawrence	NAM	5	1.267	3175	2.82	0.40	1.22	310	1570	0
19	Indus	ASIA	5	1.143	2382	2.20	0.51	5.50	1830	8240	0
20	Syr–Darya[d]	ASIA	5	1.070	1615	1.53	0.74	2.84	650	5480	40
21	Nelson	NAM	5	1.047	2045	2.00	0.56	1.06	500	3440	0
22	Orinoco	SAM	4	1.039	1970	1.93	0.58	3.01	480	5290	0
23	Murray	AUST	5	1.032	1767	1.71	0.66	1.03	260	2430	0
24	Great Artesian Basin[d]	AUST	5	0.978	1045	1.03	1.09	0.55	220	1180	70
25	Shatt el Arab	ASIA	4	0.967	2200	2.24	0.50	2.84	660	4080	0
26	Orange	AFR	5	0.944	1840	1.81	0.62	1.66	1230	3480	0
27	Huang He	ASIA	5	0.894	4168	4.41	0.26	2.93	1860	6130	0
28	Yukon	NAM	5	0.852	2716	2.90	0.39	2.93	690	6100	0
29	Senegal	AFR	4	0.847	1680	1.80	0.63	0.43	250	1070	0
30	Irharhar[b]	AFR	5	0.842	1482	1.58	0.72	1.84	500	2270	0
31	Jubba	AFR	5	0.816	1699	1.88	0.60	2.92	730	4360	0
32	Colorado (US/Mexico)	NAM	5	0.808	1808	2.01	0.56	4.50	1570	4280	0
33	Rio Grande (US/Mexico)	NAM	4	0.805	2219	2.47	0.46	3.28	1400	4240	0
34	Danube	EUR	4	0.788	2222	2.55	0.44	2.84	450	3430	0
35	Mekong	ASIA	4	0.774	3977	4.52	0.25	1.82	1030	6370	0
36	Tocantins	SAM	4	0.769	2234	2.55	0.44	1.25	390	1650	0
37	Araye[b]	AFR	4	0.742	1682	1.93	0.59	0.92	400	2880	0
38	Tarim[d]	ASIA	5	0.733	1227	1.42	0.80	7.13	2600	7460	840
39	Columbia	NAM	5	0.724	1791	2.07	0.55	4.28	1320	4300	0
40	Tafassasset[b,d]	AFR	4	0.686	1529	1.82	0.62	0.74	470	1860	260
41	Kolyma	ASIA	4	0.666	2091	2.56	0.44	2.16	490	2560	0
42	Sao Francisco	SAM	4	0.615	2212	2.82	0.40	1.46	630	1740	0
43	Amu–Darya[d]	ASIA	4	0.612	1976	2.53	0.45	5.13	1420	7110	40
44	Qattara[b]	AFR	4	0.582	1903	2.49	0.45	1.00	370	3130	0
45	Dnepr	EUR	4	0.509	1544	2.16	0.52	0.36	160	410	0
46	Dawasir[b]	ASIA	4	0.474	1435	2.09	0.54	1.51	700	2880	0
47	Don	EUR	5	0.423	1401	2.10	0.54	0.42	150	830	0
48	Colorado (Arg)	SAM	4	0.422	1750	2.32	0.49	5.57	1240	6730	0
49	Limpopo	AFR	4	0.420	1316	2.06	0.55	2.27	790	2110	0
50	Muqshin[b]	ASIA	3	0.414	1586	2.46	0.46	2.18	690	3340	0

[a] AFR = Africa; AUST = Australia; EUR = Europe; NAM = North America; SAM = South America.
[b] River system mostly non-discharging under present climate.
[c] This elevation appears to be an error in the ETOPO5 (Edwards, 1989) digital elevation model. The more reliable figure is 8848 m (Bartholemew et al., 1994, 1988).
[d] Endorheic river system, internally draining with endpoint inside continent.

through engineering works (Dynesius and Nilsson, 1994; Vörösmarty et al., 1997b) and direct anthopogenic interception and use of runoff (Vörösmarty et al., 2000b; Postel et al., 1996) requires a drainage basin perspective and knowledge of the organization of river systems.

Macro-scale river networks and their relation to source and sink areas for water and entrained particulates can also support sediment routing models (Syvitski and Morehead, 1997). The analysis of particulate fluxes for sediment and organic matter must include knowledge of the position of engineering works such as reservoirs in the cascade of river systems (Vörösmarty et al., 1997b; Stallard 1998). The concept of stream order and network organization is also at the heart of biogeochemical process models (Billen and Garnier, 1999), which eventually could be applied over the global domain. Knowledge of the organization of river systems also is important to the design and execution of river monitoring programs, particularly in the context of deriving continental and global-scale river fluxes for water and constituents (Meybeck and Ragu, 1995, 1997; Vörösmarty et al., 1997a; Grabs et al., 1996).

The STN-30p is offered free and without restriction to all interested parties who contact us at the corresponding address or at www.watsys.sr.unh.edu. STN-30p digital data will also include geo-referenced discharge time series and contributing basin attributes from the UNESCO/RivDIS version 1.1 (Vörösmarty et al. 1998a) and UNH/WMO Global Runoff Data Center (Koblenz GERMANY) run off data set maintained within the UNH-Global Hydrological Archive and Analysis System.

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